

Intervertebral Disc Disease: Physiology of CDDY

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### Abstract

Intervertebral disc disease more commonly known as (IVDD) is well known and feared disease by most short legged dog owners. This painful disease can ruin your pets life causing hefty veterinary bills and devastation. Although this progressive disease is well known most pet owners don't understand the true cause of the disease. This paper will explain the start and progression of the disease from the embryonic stage. Understanding the physiology of your dog's disease can help empower owners to advocate for the best care of their dog.

*Keywords:* IVDD, progressive disease, embryonic

### Intervertebral Disc Disease: Physiology

The physiology of Intervertebral disc disease in CDDY dogs has been a topic of debate for researchers for quite awhile. Due to lack of historical records and genetic material concrete answers on the genetic components of the disease are still being studied. The CDDY mutation causes of disease are the disease origin and process used in this paper.

#### **Fetal Physiology**

The spinal column in all members of the chordata family consists of solid bone vertebrae and the spongy gelatinous filled cartilage of the intervertebral disc. The process of IVDD occurs early during embryonic development when the fetus develops the notochord, which later on becomes the spine and is the center for the axial development of the whole body. As the fetus develops the notochord is surrounded by groups of cells called sclerotomes that turn into the bony vertebrae. These sclerotomes section off the notochord turning it into the center gelatinous part of the intervertebral discs. This gooey center is called the nucleus pulposus and is critical for the compression and flexion of the spine during movement. The start of IVDD is a dominant genetic mutation that causes chondrodystrophy (CDDY) that impacts the genes that signal growth cells during spinal cord development. Gene CDDY is an insertion mutation causing an extra copy of the fibroblast growth factor 4 (FGF4) gene to be inserted in different locations on the chromosomes. This FGF4 gene results in excess cartilage growth which impacts the formation and development of the cartilaginous endplate formation of the intervertebral discs. This results in degeneration of the nucleus pulposus which is what causes IVDD.

## **Postnatal Development**

Once the CDDY affected fetus is born there are no obvious signs of intervertebral disc issues for a period of time. While the CDDY mutation causes visibly shortened limbs with impaired growth plate cartilage growth resulting in the iconic short legs and long bodies seen in dachshunds and corgis it doesn't cause any other physical problems. Despite the changed physiology the IVDD process doesn't show visible impacts on the dogs quality of life until later stages of the disease due to its gradual widespread development. As the nucleus pulposus degrades it sends inflammation signals into the body. The body responds by calcifying the areas in an attempt to stabilize the area and reduce inflammation. However, the calcification negatively impacts the spine's ability to absorb shock and flex properly. These are critical properties of the spine as they help support the skeleton and protect the spinal cord. These limitations can cause severe back pain in the dog as their vertebrae are being pressed against a hard surface instead of the usual flex of the disc. The nerves and blood vessels that travel through the discs can be pinched which also causes severe pain or loss of function. In addition the calcification of the discs can cause the spinal cord to become impinged. Since the spinal cord is compressed the dog's motor control will be weakened or suppressed completely. Typically the hind end motor control is impacted by IVDD calcification as the discs that are more susceptible to degeneration are the lumbar area. Depending on where the spinal cord is being impinged, the dogs bladder and bowel control can be affected resulting in incontinence or the inability to relieve themselves similar to spinal cord injury patients. These limits can cause other negative processes to happen in the body as a result. Catheterizing a dog repeatedly due to its inability to urinate can cause physical trauma and result in infection.

## Comparison

In a healthy animal that is not affected by the CDDY mutation the genetic coding for the FGF4 fibroblasts is correct which results in proper anatomical development. The correct embryonic development of the notochord into the spine and intervertebral discs unimpacted by the CDDY mutation results in the dog having normal spinal anatomy with no overgrowth of cartilage or nucleus pulposus degeneration. This also means the skeletal growth is normal with no shortened limbs due to chondrodystrophy. These positives mean the dog is anatomically and physiologically sound giving it the best chance at life. While there is always a risk of the nucleus pulposus idiopathically degenerating resulting in IVDD the likely chance of occurrence in a non-CDDY dog is very low and should be of little concern.

**The physiological development of IVDD can be hard for owners to catch.** As the development timeline of the nucleus pulposus degradation and resulting calcification varies widely it can make it hard to detect on physical exam and x-ray. Since the disease has started from fetal development the best tool for CDDY dog owners is observation in an attempt to catch changes early. The rates of onset can vary widely so owners and vets are left waiting for when the disease eventually starts impacting the dog. In addition, the slow gradual spinal changes might not have any physical effect on the dog depending on the dog until the changes cause a physical failure such as pinching the spinal cord.

### **Treatments**

The current treatments for IVDD depend on the severity of the case. Dogs with slight disc degeneration are often treated with rest and steroids to reduce inflammation in and around the disc. In more serious cases such as herniation of the nucleus pulposus into the spinal cord a fenestration surgery is performed which removes the nucleus pulposus from between all of the impacted vertebrae. This releases the pressure from the degrading nucleus pulposus and prevents anymore herniations from damaging the spinal cord. These treatments aren't usually completely effective and most dogs will still have pain after.

### **Preventatives**

Ascorbic acid has been proven to help promote regeneration of the nucleus pulposus during intervertebral disc degeneration. Since ascorbic acid is vitamin C a water soluble vitamin the concern of overdosing the dog with supplementation is very low. In addition, combination glucosamine and chondroitin sulfate dietary supplementation may help prevent the further degeneration of the disc.

### **Gene editing**

With the rise of CRISPR a potential chance of being able to cut out the CDDY mutated insertion of FGF4. Further research will need to be done on this subject to determine if the removal of the FGF4 insertions would have a negative impact on the dogs. This could be a permanent solution to this negative disease that appears in a large number of dogs and a step toward breeding healthier animals.

### **Conclusion**

Intervertebral disc disease caused by the chondrodystrophy gene mutation CDDY is a progressive and devastating disease. It causes the degeneration of the gelatinous center of the intervertebral discs resulting in calcification, herniation, and intense pain due to the functional part of the discs being impeded. The CDDY mutations aren't fully understood yet, but with further research and technological developments a treatment and solution may be created.

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